

Interferometric Study of the Normal and Anomalous Zeeman Effects in Cadmium

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Abstract

The Zeeman effect was discovered in 1896 by the physicist of the same name. This phenomenon consists of a lifting of the degeneracy of the energy levels of an atom in the presence of an external magnetic field. It is possible to observe the consequences of such a phenomenon by examining the emission spectrum of this atom, in which certain spectral lines are shifted. To measure this shift (on the order of 10^{-11} m in wavelength), we will use a Fabry-Pérot interferometer. The purpose of this practical session is therefore to study the normal and anomalous Zeeman effects, the polarization of these lines, and to measure the value of the Bohr magneton.

I. ELEMENTS OF THEORY

A. Quantum Numbers

We introduce the following quantum numbers:

L : orbital quantum number

S : total spin quantum number

J : total angular momentum quantum number

M_J : magnetic quantum number associated with the total angular momentum, $M_J \in \{-J, \dots, J\}$

These numbers will be useful for characterizing the energy levels according to atomic spectroscopy notation: $^{2S+1}L_J$

B. Hamiltonian in the Presence of a Magnetic Field

In the absence of a magnetic field, both the electrostatic term and the spin-orbit coupling must be taken into account:

$$\hat{H}_0 = \hat{H}_{\text{electrostatic}} + \hat{H}_{\text{coupling}} \quad (1)$$

In the presence of a magnetic field B , the Hamiltonian becomes:

$$\hat{H} = \hat{H}_0 + \hat{V} \quad (2)$$

with the Zeeman Hamiltonian:

$$\hat{V} = \frac{-e}{2m} (\hat{L}_z + g_e \hat{S}_z) B \quad (3)$$

This modification of the Hamiltonian caused by the addition of a magnetic field induces a lifting of the degeneracy of the energy levels into $2J + 1$ distinct, equidistant levels.

It is then possible to express the energy variation between 2 consecutive levels as a function of the intensity of B :

$$\Delta E = h\Delta\nu = B|\mu_B|(g_{J'}M_{J'} - g_JM_J) \quad (4)$$

with

$$\mu_B = \left(\frac{e\hbar}{2m} \right) \approx -9.27 \times 10^{-24} \text{ J} \cdot \text{T}^{-1}$$

being the Bohr magneton and g_J the Landé g-factor, whose expression is as follows:

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} \quad (5)$$

We will first focus on the normal Zeeman effect, with the $^1D_2(L=2, S=0, J=2) \rightarrow ^1P_1(L=1, S=0, J=1)$ transition of the Cadmium atom (Fig. 1), whose Landé g-factors are 1 for both levels. Next, we will study the anomalous Zeeman effect, corresponding to the $^3S_1(L=0, S=1, J=1) \rightarrow ^3P_2(L=1, S=1, J=2)$ transition, whose Landé g-factors are 2 and $\frac{3}{2}$ respectively.

C. Polarization of Light

The observed transitions are polarized according to the variation in ΔM_J . There is indeed a $\Delta M_J = 0$ transition, called the π transition because it is linearly polarized parallel to the B field. The $\Delta M_J = \pm 1$ transitions are called σ transitions because they are elliptically polarized for any arbitrary direction of observation.

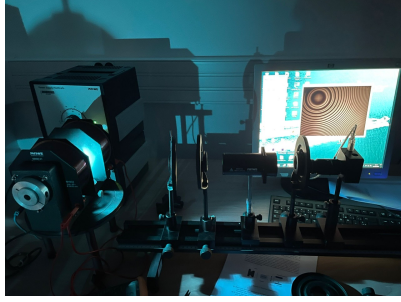
D. Fabry-Pérot Interferometer

Rings (or fringes) are conventionally observed on a Fabry-Pérot interferometer. However, the Zeeman effect causes a spectral shift $\Delta\lambda$ which results in the subdivision of the various observed rings. If we denote R_λ as the radius of the central ring and $R_{\lambda 1}$ and $R_{\lambda 2}$ as the radii of the inner and outer rings respectively (which correspond to the lifting of degeneracy caused by B); it is possible to arrive at the following relation (to the 1st order):

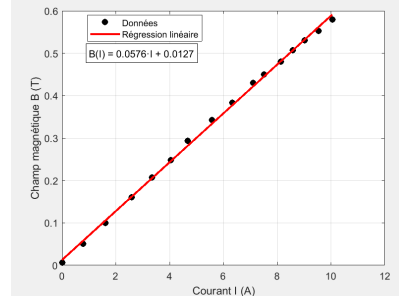
$$\frac{R_\lambda^2}{R_{\lambda 2}^2 - R_{\lambda 1}^2} \approx \frac{\lambda^2}{4d\Delta\lambda}(p - 1 + \epsilon) \quad (6)$$

where d is the distance between the 2 Fabry-Pérot mirrors ($d = 0.003$ m), p is the interference order, and λ is the wavelength of the studied Cadmium line.

In the following sections, we will primarily use equations (4) and (6) to analyze our results.



(a)



(b) Relationship between the field B and the current I

Figure 2: Experimental settings

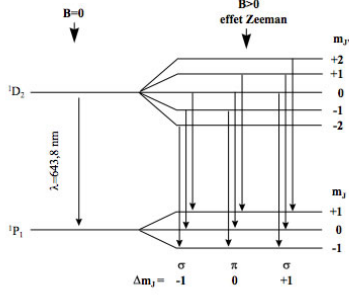


Figure 2: Effet Zeeman sur la transition $1D_2 \rightarrow 1P_1$ (raie rouge du Cadmium à 643,8 nm)

Figure 1: Illustration of the different possible transitions

II. EXPERIMENTAL SETUP

A. Description of the Setup

We have a Cadmium lamp placed in the center of a coil. We are careful not to exceed a current of 10 A in the coil in order not to damage it.

Placed successively on an optical bench are: a diaphragm, the Fabry-Pérot interferometer, a lens with focal length $f = 10$ cm, and the camera (Fig. 2a).

B. Magnetic Field Calibration

In order to analyze our results, it is necessary to calibrate the magnetic field of the coil. To do this, the cadmium lamp is removed from its holder and $B(I)$ is plotted using the Gaussmeter. (Fig. 2b)

III. STUDY OF THE NORMAL ZEEMAN EFFECT

A. Initial Observations

We first look at the polarization of the different transitions. When the magnetic field is orthogonal to the radiation, the π transition appears to us as linear in the direction of B. On the other hand, the σ transitions appear as linear, orthogonal to B, and shifted by π from each other. This is why we observe an extinction of the outer rings (corresponding to the σ transitions) for a polarization angle of 0° (Fig. 3a).

It is possible to observe all transitions (but with a reduced luminous intensity) for an angle of 10° , for example (Fig. 3b).

Conversely, when the magnetic field is parallel to the beam, the σ^+ and σ^- transitions appear circular (left and right, respectively) and the π transition does not appear. This is why, for any polarization angle, the outer rings are visible and the central rings have a weak intensity (Fig 3c). One should, however, expect a total extinction of the central rings; this could be explained by a misalignment between the optical bench and the coil + cadmium lamp assembly.

B. Calculation of the Bohr Magneton

The objective is to calculate the value of μ_B by measuring the splitting of the Zeeman levels. To do this, let's return to equation (6):

$$\frac{R_{\lambda}^2}{R_{\lambda_2}^2 - R_{\lambda_1}^2} \approx \frac{\lambda^2}{4d\Delta\lambda}(p-1+\epsilon) \quad (7)$$

By measuring the 3 radii at different orders p , we can plot the curve $\frac{R_{\lambda}^2}{R_{\lambda_2}^2 - R_{\lambda_1}^2} = f(p)$ whose slope provides access to $\Delta\lambda$.

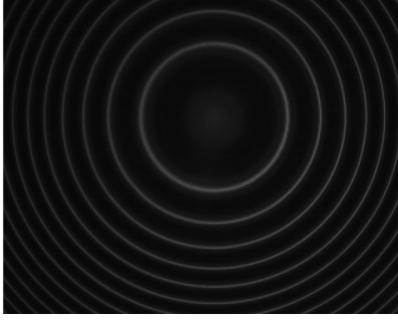
Furthermore, relation (4) allows us to deduce:

$$\Delta\lambda = \mu_B \left(\frac{\lambda^2}{hc} \right) B \quad (8)$$

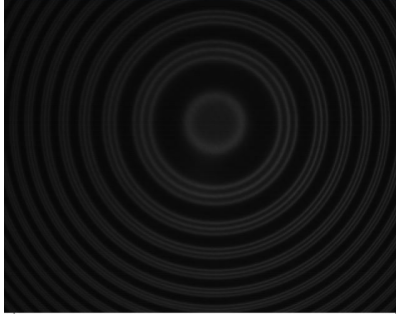
It then suffices to calculate the values of $\Delta\lambda$ for different values of B to gain access to μ_B .

C. Experimental Method

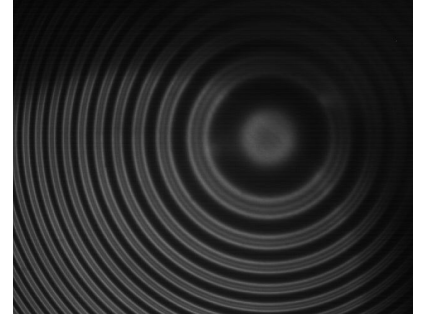
To perform the measurement of the Bohr magneton, we process the image taken by the camera using a MATLAB script that measures the pixel intensity as a function of the distance to the center of the rings (Fig. 4a). In addition, we operate at a relatively high B value (corresponding to a current between 7 A and 10 A). Indeed, since ΔE is proportional to the intensity of B, the spectral shift induced by the magnetic field will be greater for a strong field (Fig.



(a) Rings in perpendicular geometry, polarization angle 0° , $I=8.06$ A

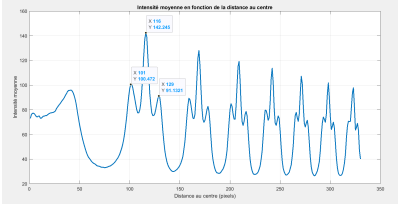


(b) Rings in perpendicular geometry, polarization angle 10° , $I=8.06$ A

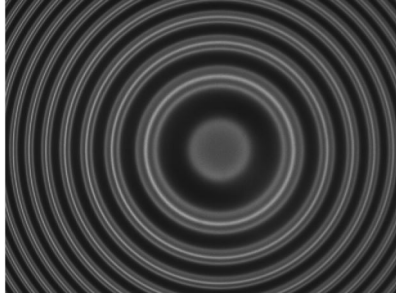


(c) Rings in parallel geometry, arbitrary polarization angle, $I=7.78$ A

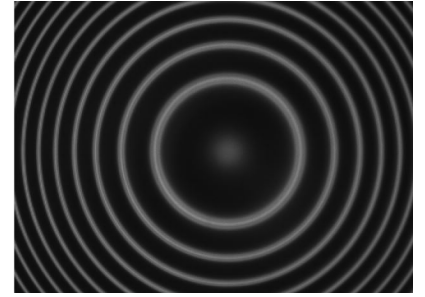
Figure 3: Observation of the rings under different polarization angles.



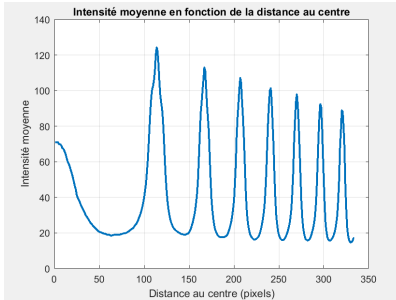
(a) Measurement of the radii using the MATLAB script. Here, measurement of the central, inner, and outer radius at order 1 for $I=9.57$ A.



(b) Observation of the rings for $I=9.57$ A.



(c) Observation of the rings for $I=4.11$ A. The inner and outer rings are not clearly visible due to the small spectral shift.



(d) Script output for the previous image. The measurement of the radii R_λ , $R_{\lambda 1}$, and $R_{\lambda 2}$ is not achievable.

Figure 4: Results of the ring image analysis at different intensities.

4b). Conversely, it is difficult to observe this shift for fields that are too weak (Fig. 4c, 4d).

D. Experimental Results

The results for the determination of $\Delta\lambda$ are presented in Fig. 5. Knowing the relationship between B and I through the performed calibration, it is possible to carry out a linear regression of $\Delta\lambda$ as a function of B (Fig. 5h) and deduce the value of μ_B .

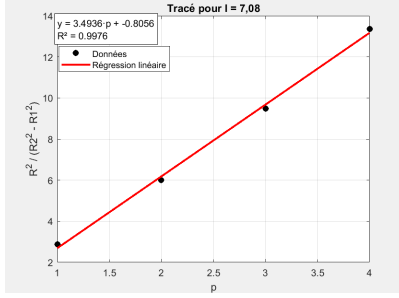
With $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$, $c = 3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ and $\lambda = 622 \text{ nm}$ for the red line of Cadmium, we obtain $\mu_B =$

$$15.14 \times 10^{-24} \text{ J} \cdot \text{T}^{-1}.$$

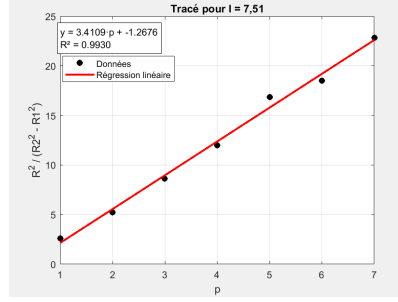
IV. STUDY OF THE ANOMALOUS ZEEMAN EFFECT

A. Electronic Structure

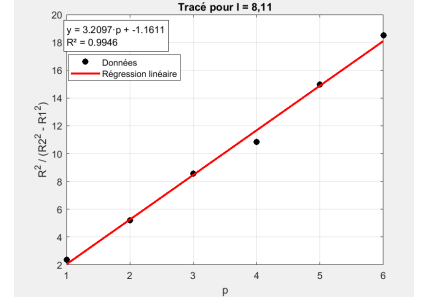
We now focus on the green line of Cadmium, i.e., the $^3S_1(L = 0, S = 1, J = 1) \rightarrow ^3P_2(L = 1, S = 1, J = 2)$ transition with Landé g-factors equal to 2 and $\frac{3}{2}$ respectively (equation 5). The Landé g-factors for the 2 energy levels are different, so the energy spacing of the sub-levels is



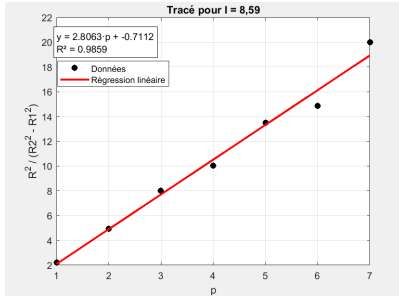
(a) Measurements for I=7.08 A



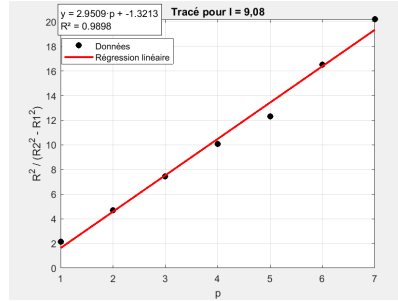
(b) Measurements for I=7.51 A



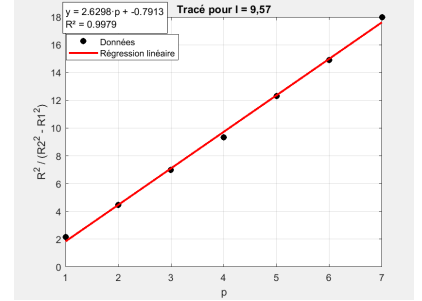
(c) Measurements for I=8.11 A



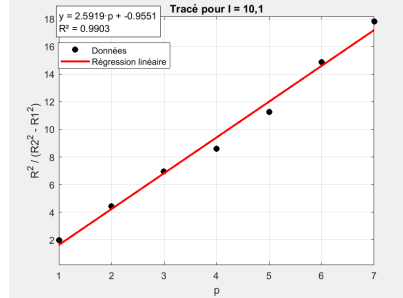
(d) Measurements for I=8.59 A



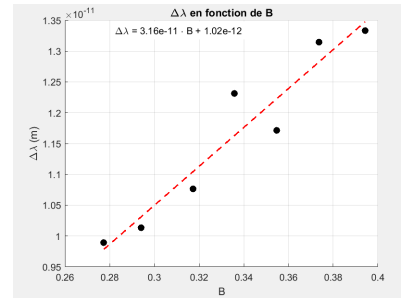
(e) Measurements for I=9.08 A



(f) Measurements for I=9.57 A



(g) Measurements for I=10.1 A



(h) Linear regression of the spectral shift as a function of the field

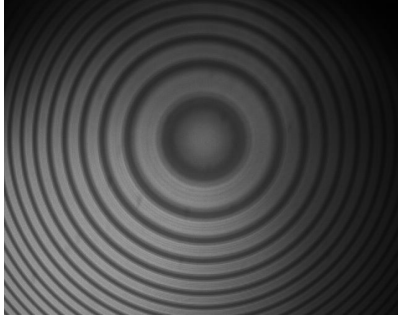
Figure 5: Overall experimental results for the determination of $\Delta\lambda$ and μ_B

different for the upper level ($\mu_B B g_{J'}$) and the lower level ($\mu_B B g_J$). This is why in the anomalous Zeeman effect, we observe a subdivision of the rings into more than 3 (unlike the normal Zeeman effect). We know this is a $^3S_1(L=0, S=1, J=1) \rightarrow ^3P_2(L=1, S=1, J=2)$ transition, so $M_J \in \{-1, 0, 1\}$ and $M_{J'} \in \{-2, -1, 0, 1, 2\}$, meaning there are 15 transitions in total. Only 9 are allowed according to the selection rules. Therefore, we should be able to see 9 lines for the anomalous Zeeman effect.

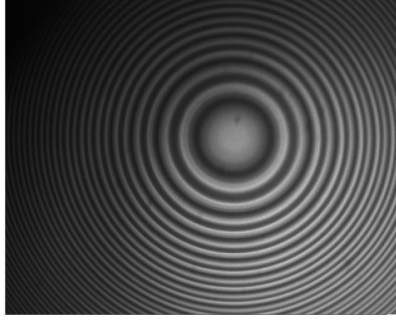
B. Observations of the Rings

The observed subdivision (Fig. 6a) can sometimes make identifying the rings difficult. Indeed, due to the multiplicity of these rings, the orders can overlap, particularly in the pres-

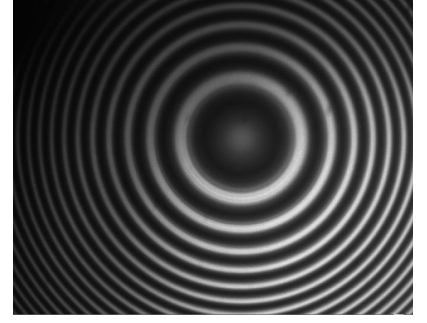
ence of a relatively strong B field. Unlike the normal Zeeman effect, we observe "polarization packets" (or groups). Indeed, for the normal Zeeman effect, there were π and σ transitions that corresponded to a single energy sub-level. They could therefore be extinguished one by one by adjusting the polarization. However, for the anomalous Zeeman effect, several sub-levels share the same polarization, leading to the observation of "packets" that extinguish simultaneously under the effect of a polarizer. (Fig. 6b, 6c, 6d, 6e, 6f)



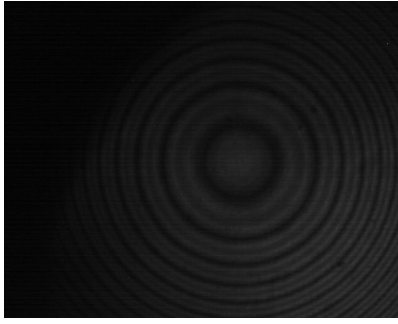
(a) Multiplicity of rings in the anomalous Zeeman effect



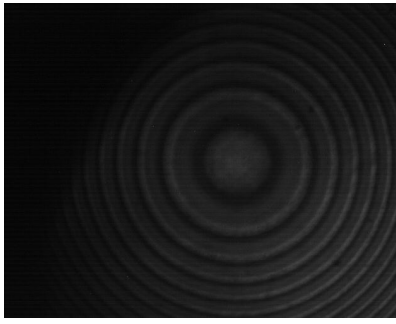
(b) Perpendicular geometry, polarization 0°, I=8.12 A



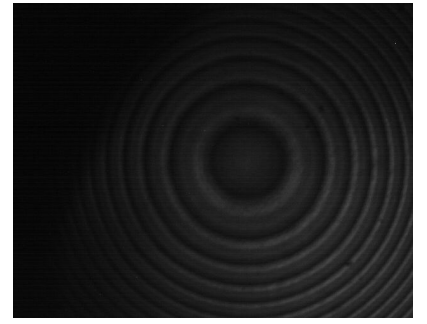
(c) Perpendicular geometry, polarization 90°, I=8.12 A



(d) Parallel geometry, polarization 0°, I=8.12 A



(e) Parallel geometry, right 45° polarization, I=8.12 A



(f) Parallel geometry, left 45° polarization, I=8.12 A

Figure 6: Observation of the rings with the anomalous Zeeman effect

V. DISCUSSION

A. Experimental Value of the Bohr Magneton

We find a value of $\mu_B = 15.14 \times 10^{-24} \text{ J} \cdot \text{T}^{-1}$, which represents a relative error of 63 %. The sources of uncertainty are multiple. Firstly, in order to observe a sufficient shift in the rings, a relatively high current must be set (in this case, at least 7 A). This restricts the measurement range, meaning the final regression of $\Delta\lambda$ as a function of B will contain fewer experimental points (Fig. 5h). Moreover, for this same reason, the measurements taken for currents around 7 A are less precise, as it is not possible to measure the radii for higher orders (Fig. 7).

If we calculate the value of μ_B exclusively for higher currents where the script is more accurate (i.e., by removing the first 3 measurements and keeping $I = 8.59 \text{ A}, 9.08 \text{ A}, 9.57 \text{ A}, 10.1 \text{ A}$), we then find $\mu_B = 11.10 \times 10^{-24} \text{ J} \cdot \text{T}^{-1}$, giving a relative error of 19.7 %. This tends to confirm what was stated above.

Aside from possible camera adjustments (increasing exposure time, gain, and adjusting contrast), a potential solution is to improve the provided MATLAB script. Indeed, in the majority of the photos taken, distinguishing the rings

was easier in the lower-left quadrant, where their numbers were also higher (this asymmetry was due to the optical bench alignment and the differing heights of the various instrument stands). Therefore, executing the script along a more restricted, specific direction (a single line rather than an angular sector) might have improved measurement quality, even for lower currents.

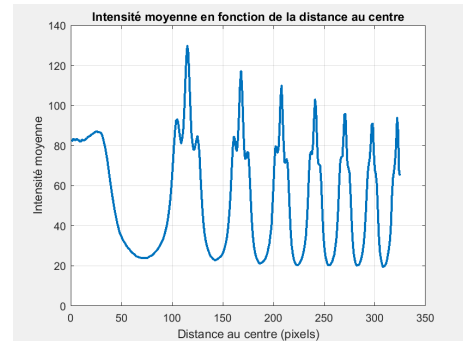


Figure 7: Script output for I=7.08 A; beyond $p = 5$, measuring $R_{\lambda 1}$ and $R_{\lambda 2}$ is no longer possible

B. Observation of the Anomalous Zeeman Effect

In the case of the anomalous Zeeman effect, the overlapping of orders at strong fields makes interpretation challeng-

ing. However, for the same reasons as in the normal Zeeman effect, we try to determine $\Delta\lambda$, and therefore μ_B , at strong fields. The uncertainty in measuring μ_B is thus increased.

Furthermore, the high multiplicity of the rings makes identifying the different radii difficult (Fig. 8). The measurement would thus be significantly less precise than in the normal Zeeman effect. Therefore, we could, at best, only calculate an order of magnitude for μ_B .

When observing the rings, it was also harder to determine the polarizer's effect in the parallel configuration because the luminous intensity of the rings was significantly diminished.

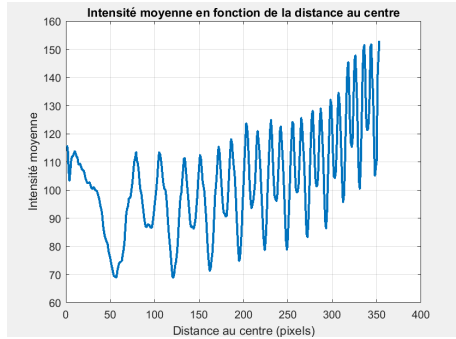


Figure 8: Script output in the anomalous Zeeman effect for $I=8.72$ A; identifying the sub-levels is no longer possible